

MARTIN LAFARGUE

Ph.D. researcher in Computational Aerodynamics | CFD | Unsteady Aerodynamics | Aeroacoustics | HPC

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EDUCATION

PHD IN OCEAN ENGINEERING | FLORIDA ATLANTIC UNIVERSITY

2023-PRESENT

Research specialization: Computational Aerodynamics, Unsteady Aerodynamics & Aeroacoustics

- Research on an innovative technique of stall mitigation using interacting airfoils — OpenFOAM, Overset Mesh, Aerodynamics, Stall, HPC [CURRENT]
- *Aeroacoustic inversion of noise generated by the interaction of an unsteady gust with a rotor, Lafargue&Verma, Journal of Theoretical and Computational Aeroacoustics (August 2026)* — Analytical Aeroacoustics, MATLAB, Inversion Techniques, Wavelet Transform, Deep Learning
- Teaching Assistant of Computational Applications in Engineering 1 (MATLAB) [SPRING 2026]
- Simulations of flow in a pipe for a sprinkler from Krain Company [SPRING 2026]

MASTER OF AEROSPACE ENGINEERING | UNIVERSITY OF CINCINNATI

AUGUST 2022

- Double degree program with ENSEIRB-MATMECA
- Capstone Research Project: "Ionic Breeze" — Fluent Numerical Study to determine if combining ionic wind and aerodynamic flow control concept would enhance the lift on a wing [FALL 2021]

ENGINEERING DIPLOMA (FRANCE) | ENSEIRB-MATMECA

2019 – 2022

- Double degree program with University of Cincinnati during the fifth year of higher study
- Advanced modeling methods and numerical simulation for mechanical and fluid mechanics applications
- Fifth, fourth and third year of higher study at ENSEIRB-MATMECA: Bordeaux Graduate School of Engineering, Mathematics and Computer Modeling Department
- ACADEMIC GROUP PROJECT: 2D modeling of composites mechanical behavior at strand scale | C++, linear elasticity, meshing, image acquisition and processing, 6 persons, 80h of work | 2020–2021|
- ACADEMIC GROUP PROJECT: Modeled, coded, and conducted analysis to study knee joint articulation during the human walk | FORTRAN 90, mechanics, mathematics, 4 persons, 60h | 2019–2020|

UNDERGRADUATE STUDIES | LA PRÉPA DES INP, BORDEAUX

2017 – 2019

- A 2-year period of intensive courses in mathematics, physics, mechanics, chemistry, and biology in preparation for entrance in French engineering schools of the INP group

EXPERIENCES

AEROSAPIEN TECHNOLOGIES — Aerodynamics Engineer Intern | Daytona Beach, FL

April- September 2022

- Performed CFD simulation analysis of lift and drag values of a box-wing aircraft concept as a function of angle of attack using STAR-CCM+ and Pointwise Meshing
- Researched and prepared technical documents
- Conducted 2D & 3D, structured and unstructured meshing
- Participated in team discussions to enhance new techniques and resolve challenges

DIRECTION GÉNÉRALE DE L'ARMEMENT — CFD Intern | Saint-Médard-en-Jalles, France

June–August 2021

- Fluent numerical study to determine optimal conditions to avoid hot gas blowbacks for a thruster

SONOVISION ORTEC GROUP — Intern | Colomiers, France

April–June 2019

- Redaction of documents describing tasks for different roles to enhance the team's interoperability

SKILLS

ENGINEERING & ANALYSIS

Fluid mechanics · Unsteady Aerodynamics · Aeroacoustics · Wavelet Analysis. Fourier Analysis. SVD · PCA · Physics · Mathematics

COMPUTING

CFD (OpenFOAM, ANSYS Fluent, StarCCM+, Overset Meshes), FEA (Abaqus), Pointwise Meshing, Paraview C++, FORTRAN, Python, MATLAB, HPC, Deep Learning (Tensorflow/Keras, Neural Networks), Git, GitHub

LANGUAGES

English: Very good user (IELTS Level 8, Dec 2020)
Spanish: Intermediate
French: Native Speaker

EXTRA-CURRICULAR ACTIVITIES

- Football (soccer) in club since age 6; volunteer coach of under-7 category from September 2015 to June 2017
- Free practice cycling — participated in 3-day cyclo-sport race "Campilaro" in 2018, 2019, 2021 & 2023 in the Pyrenees
- 1st Place, Men's 25-29, Sprint Winter Triathlon, USA Triathlon Winter National Championships, Breckenridge, CO (Feb. 2025)
- 3rd Place, Men's 25-29 (20/124 overall), Tri KB International Triathlon, Key Biscayne, FL (Sept. 2024)
- Reading & Writing: created a sports website publishing articles at www.sportsailevasion.com; creative writing